# Vitor Albert

Software Engineer | AI/LLM Systems & Backend

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I build AI features that have to work in front of real users, and the backend that keeps them running.

## Experience

**Founder & Lead Engineer | AfterLands** Dec 2022 to Present

*Minecraft MMORPG network*

- Built CORTEX, the community's Discord assistant. It answers player questions from our own docs using RAG on Qdrant, mixing vector and keyword scores and normalizing Brazilian slang before searching. About 62k lines of Python and 1,600 tests.
- Any AI call that can affect a player (moderation, auto-answers, what the bot remembers) goes through a decision layer I wrote. New uses start in shadow mode, fall back to fixed rules when the model times out, strip personal data first and treat user text as untrusted. Each decision is logged for review.
- Wrote a test harness that lets AI agents check features inside the game on their own: they read a checklist, run console commands and SQL, and drive a real player bot (mineflayer). It can only reach the test servers. 89 checklists passed so far.
- Run most development through AI agents with a spec, plan and in-game check for every change. That's how one engineer shipped 5,800 commits in seven months on a 34-module Java 21 codebase with 15,800 unit tests and CI on GitHub Actions.
- I run the infrastructure myself: 20+ game servers spread across several dedicated machines (OVH, Hetzner), mostly Ubuntu, managed through Pterodactyl. I handled the security setup end to end, from SSH hardening and firewall rules to access protocols, plus daily backups to Backblaze B2. Prometheus and Grafana watch everything and post alerts and a daily backup report to Discord.
- Designed AfterCore, the shared library under 30+ plugins: async MariaDB access, Redis messaging and distributed locks, circuit breakers and health checks. I also run and tune MariaDB and Redis in production. The whole stack targets 500+ players online at 20 TPS. On top of it I built what players see: NPCs that replay recorded motion and turn to face you, camera cinematics, 2,000+ block animations and Crowdin-synced translations.

**Freelance Software Engineer & Game Designer** 2022 to Present

- Ran the tech for a live High School event with YouTuber Savvy Soggy. 350+ players were on at once during filming; I load-tested it beforehand and it didn't crash once.
- Designed TabiQuest, a Cobblemon RPG server: the story campaign, branching dialogue, difficulty curve and the economy.
- Wrote the design document for Dinosaurs: Lost Depths, a Minecraft Marketplace add-on for Panda Studios.
- Set up and secured dedicated servers and VPSs for clients on OVH and Hetzner, from first install to monitoring.
- Other clients: Mia Studios and Eco Studios.

## Open Source

- **IGLanguages**: a Spigot plugin that lets each player pick their language, with Crowdin integration. About 875 downloads across Spigot and Modrinth, 224 players at its peak (bStats).

## Skills

**AI:** RAG, embeddings, Qdrant, Gemini, OpenRouter, evals, prompt-injection defense, Claude Code
**Languages:** Java 21, Python, TypeScript, SQL
**Backend:** Redis, MariaDB/MySQL, SQLAlchemy, Alembic, HikariCP, JUnit 5, Mockito, pytest
**Infra:** Linux (Ubuntu), Pterodactyl, Prometheus, Grafana, Backblaze B2, Docker, GitHub Actions, Cloudflare, OVH, Hetzner
**Games:** Spigot/Paper, Velocity, PacketEvents, game design documents

## Education & Languages

Game Design: Art and Concepts, CalArts (Coursera), Oct 2025
Information Systems coursework, UFG (paused)
Portuguese (native), English (advanced)